

Credit Score classification Project

Objective:

Build and train a classification model to predict credit scores for customers using **Azure Machine Learning Designer**, and deploy the model for real-time predictions.

Steps Performed:

Create Resource group

[Home](#) > [Resource Manager](#) | [Resource groups](#) >

Create a resource group ...

×

Basics Tags Review + create

[Automation Link](#)

Basics

Subscription	Azure subscription 1
Resource group name	rg_azure
Region	East US

Tags

None

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Azure ML → Create workspace

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Azure Machine Learning ...

×

Create a machine learning workspace

✓ Validation passed

Basics Inbound Access Outbound Access Encryption Identity Tags Review + create

Subscription	Azure subscription 1
Resource group	rg_azure
Region	East US
Name	ws_azure
Storage account	(new) wsazure5284963786
Key vault	(new) wsazure9426880684
Application insights	(new) wsazure9876887849
Container registry	None

Connectivity method	Enable public access from all networks
Network isolation	Public

Encryption type	Microsoft-managed keys
-----------------	------------------------

Identity: none

System assigned

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[Download a template for automation](#)

Home > Microsoft.MachineLearningServices | Overview

Deployment

Search

Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

✓ Your deployment is complete

Deployment name : Microsoft.MachineLearningServices
Subscription : Azure subscription 1
Resource group : rg_azure

Start time : 2/2/2026, 7:38:31 AM
Correlation ID : 532197fd-f752-4158-8532-d77b039d44e8

> Deployment details

> Next steps

Go to resource

Deployment succeeded

Deployment 'Microsoft.MachineLearningServices' to resource group 'rg_azure' was successful.

Go to resource Go to resource group

Cost management

Get notified to stay within your budget and prevent unexpected charges on your bill.

Set up cost alerts >

Microsoft Defender for Cloud

Secure your apps and infrastructure

Go to Microsoft Defender for Cloud >

Free Microsoft tutorials

Start learning today >

Work with an expert

Azure experts are service provider partners who can help manage your assets on Azure and be your first line of support.

Find an Azure expert >

Add or remove favorites by pressing Ctrl+Shift+F

Launch ML Studio

Microsoft Azure Upgrade

Search resources, services, and docs (G+)

Copilot

santhoshc1902@gmail.com

Home > Microsoft.MachineLearningServices | Overview

ws_azure

Azure Machine Learning workspace

Suggest a workload for this ML workspace

Search

Download config.json Delete

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Events

Settings

Monitoring

Automation

Support + troubleshooting

Essentials

Resource group : rg_azure

Location : East US

Subscription : Azure subscription 1

Storage : wsazure5284963786

Provisioning State : Succeeded

Studio web URL : https://ml.azure.com?tid=c064b21c-06dc-409b-a09e-c1e664eba...

Container Registry : ...

Key Vault : wsazure9426880684

Application Insights : wsazure9876887849

MLflow tracking URI : azureml://eastus.api.azureml.ms/mlflow/v1.0/subscriptions/04a2...

JSON View

Work with your models in Azure Machine Learning Studio

The Azure Machine Learning Studio is a web app where you can build, train, test, and deploy ML models. Launch it now to start exploring, or learn more about the Azure Machine Learning studio

Launch studio

Add or remove favorites by pressing Ctrl+Shift+F

Create Compute

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Create compute instance

1 Required settings

2 Scheduling optional

3 Security optional

4 Applications optional

5 Tags optional

6 Review

Virtual machine type

CPU GPU

Virtual machine size

Select from recommended options Select from all options

Name ↑	Category	Workload types	Available quota	Cost
Standard_DS11_v2 2 cores, 14GB RAM, 28GB storage	Memory optimized	Development on Notebooks (or other IDE) and light weight testing	6 cores	\$0.18/hr
Standard_DS3_v2 4 cores, 14GB RAM, 28GB storage	General purpose	Classical ML model training on small datasets	6 cores	\$0.29/hr
Standard_E4ds_v4 4 cores, 32GB RAM, 150GB storage	Memory optimized	Data manipulation and training on medium-sized datasets (1-10GB)	4 cores	\$0.29/hr

Review + Create Back Next Cancel

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Default Directory > ws_azure > Compute

Compute

The "Kubernetes clusters" tab is now where you can access previous versions of "inference clusters" (also known as "AKS clusters") and "attached Kubernetes" compute types along with any previously created compute targets using those types. [Learn more](#) about Kubernetes clusters.

Compute instances Compute clusters Kubernetes clusters Attached computes Serverless instances

Choose from a selection of CPU or GPU instances preconfigured with popular tools such as VS Code, JupyterLab, Jupyter, and RStudio, ML packages, deep learning frameworks, and GPU drivers. [Learn more about compute instances](#)

+ New Refresh Start Stop Restart Schedule and idle shutdown Delete Reset view View quota

Search

Filter Columns

Name	☆	State	Idle shutdown	Applications	Size
compute-ml1		Running	1 hour	JupyterLab Jupyter VS Code (Web) ...	Standard

Data Load

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Create data asset

Data type

Data source

Set the name and type for your data asset

Name *
credit-score

Description
Designer model for credit score classification

Type *
Tabular

Use cases for data types

When should I use File type?

The File type is recommended in most scenarios when you are working with a single data file of any type (including tabular data). This type allows you to specify a file location by URI in a storage location on your local computer, an attached Datastore, blob/ADLS storage, or a publicly available http(s) location. There are many types of supported URIs. In the Azure Machine Learning CLI v2 or Python SDK v2, this data type is called uri_file. [Learn more about the uri_file type](#)

When should I use Folder type?

The Folder type has all the same capabilities and uses...

Back Next Cancel

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Create data asset

Data type

Data source

Destination storage type

File or folder selection

Settings

Schema

Review

Review the settings for your data asset and make any changes as needed.

Data type

Name
credit-score

Description
Designer model for credit score classification

Type
tabular

Data source

Type
Local

File selection

Upload path

Schema

Month	String
Age	String
Occupation	String
Annual_Income	String
Monthly_Inhand_Salary	Decimal

(showing 5 of 29 columns)

Back Create Cancel

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Linked Services

Default Directory > ws_azure > Data

Data

Data assets Datasets Data import (PREVIEW) Data connections (PREVIEW)

Data assets are immutable references to your data that can be created from datastores, local files, public URLs, or Open Datasets. Data assets created with AzureML v2 APIs cannot be deleted, but you can up-version or archive them for easy referencing and reuse in machine learning tasks. Deleting data assets created with v1 APIs will permanently delete the data asset and all metadata. [Learn more about data assets](#)

+ Create Refresh Archive Reset view Show latest version only Include archived View my data

Search Filter Columns

Name	☆	Ve...	Data source	Created on ↓	Modified on	Type
credit-score		1	workspaceblobstore	Feb 2, 2026 9:07 AM	Feb 2, 2026 9:07 AM	Table
insurance		1	workspaceblobstore	Feb 2, 2026 7:49 AM	Feb 2, 2026 7:49 AM	Table

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Azure Machine Learning Designer

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Default Directory > ws_azure > Designer > Authoring

Undo Redo Validate Show lineage Clone AutoSave Configure & Submit

Pipeline-Created-on-02-02-2026

Save Pipeline interface

credit-score credit-score V 1

Data output Dataset

Clean Missing Data clean_missing_data

Cleaned dataset Cleaning tran... Dataset

Remove Duplicate Rows remove_duplicate_rows

Result Dataset

Split Data split_data

Multiclass Decision Forest multiclass_decision_forest

Untrained model

Untrained mod... Dataset

Train Model train_model

Trained model

Navigator 110%

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Undo Redo Validate Show lineage Clone AutoSave Configure & Submit

Pipeline-Created-on-02-02-2026

Save Pipeline interface

Train Model train_model

Trained model

Split Data split_data

Results-datas... Results-datas...

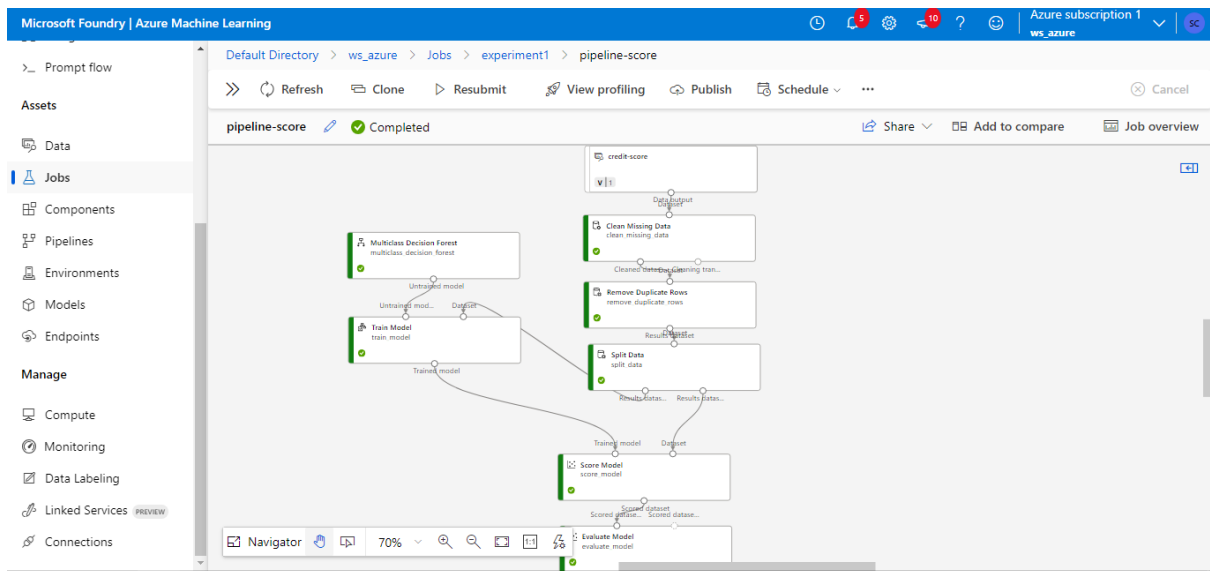
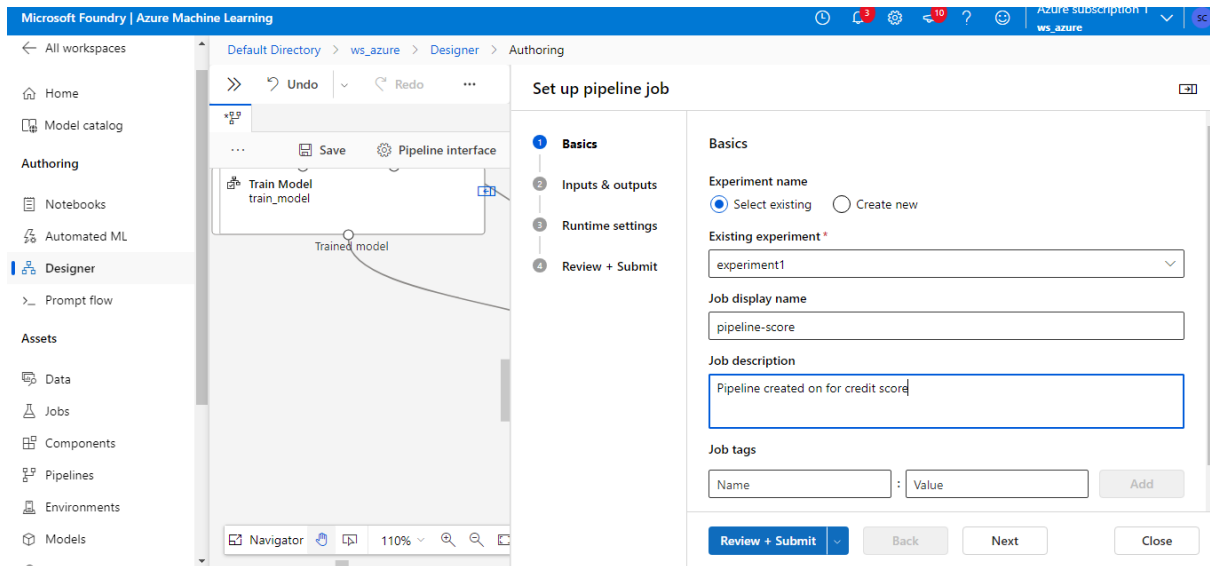
Trained model Dataset

Score Model score_model

Scored dataset Scored dataset... Scored dataset...

Evaluate Model evaluate_model

Navigator 110%



Navigate to Train model and Register the model

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Default Directory > ws_azure > Jobs > experiment1 > pipeline-score

pipeline-score ☒ Completed

Share Add to compare Job overview

Train Model

Overview Parameters Outputs + logs Metrics Child jobs Images

Properties

Status: ☒ Completed

Created on: Feb 2, 2026 9:37 AM

Start time: Feb 2, 2026 9:37 AM

Duration: 18.48s

Compute duration: 18.48s

Name: bfd5e4de-4455-45aa-9399-fbd4172bfe23

Inputs

Input name: Untrained_model
Dataset: d2caac7e-84af-404b-b111-4718c041e901

Input name: Dataset
Dataset: 5c746bc4-e2e4-4bca-8ac6-3664f18a263b

Outputs

Output name: Trained_model
Dataset: ae95bef8-85cc-49a8-812b-ebf0885fba4f

Tags

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Default Directory > ws_azure > Jobs > experiment1 > pipeline-score

pipeline-score Completed

Train Model

Overview Parameters **Outputs + logs** Metrics Child jobs Images

Refresh + Register model Debug and monitor

Data outputs [Show data outputs](#)

Other outputs

Select a File to Preview

Register model from a job output

1 Select job
2 **Select output**
3 Model settings
4 Review

Select output
Specify the corresponding output to register the model.

Model type *

Job output *

7 files selected

File name

- ☐ trained_model_outputs/_meta.yaml
- ☐ trained_model_outputs/_samples.json
- ☐ trained_model_outputs/_schema.json

Back Next Cancel

Default Directory > ws_azure > Jobs > experiment1 > pipeline-score

pipeline-score Completed

Train Model

Overview Parameters **Outputs + logs** Metrics Child jobs Images

Refresh + Register model Debug and monitor

Success: Model 'model-score:1' registered successfully. [Click here to go to this model](#)

Data outputs [Show data outputs](#)

Other outputs

Deploy model

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Prompt flow

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Default Directory > ws_azure > Models > model-score:1

model-score:1

Details Versions Artifacts Endpoints Jobs Data Feature sets Responsible AI

Refresh

Archive

Use this model

Download all

Share model

Attributes

Name

model-score

Version

1

Created on

Feb 2, 2026 10:31 AM

Created by

Santhosh Chinnasamy

Type

CUSTOM

Real-time endpoint

Deploy the model using the real-time endpoint wizard

Batch endpoint

Deploy the model using the batch endpoint wizard

Web service

Deploy to a web service (only for models based on frameworks)

Tags

No tags

Properties

azureml.datastoreid : /subscriptions/04a228dc-239f-4e96-a50c-3e0dfecbafa5/resourceGroups/rg_azure/providers/Microsoft.MachineLearningServices/workspaces/ws_azure/datastores/workspaceartifactstore

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Deploy model-score:1

Endpoint

Model

Deployment

Code + environment

Compute

Traffic

Review

Select endpoint

An endpoint is used to deploy and score your models. [Learn more](#)

Endpoint

New

Existing

Endpoint name *

endpoint-score

Description

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Cancel

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🕒 7 🛠️ 10 ? 😊

Azure subscription 1
ws_azure

Deploy model-score:1

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d3105e15-d458-46f8-9cde-6383329fd3e6 | Overview

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Redeploy

Download

Refresh

Overview

The resource write operation failed to complete successfully, because it reached terminal provisioning state 'Failed'. Click here for more details.

Your deployment failed

Deployment name : d3105e15-d458-46f8-9cde-6383329fd3e6
Subscription : Azure subscription 1
Resource group : rg_azure

Start time : 2/2/2026, 10:31
Correlation ID : 3171dae9-b46f-4b4d-8b4d-3171dae9-b46f

Deployment details

Resource	Type	Status
ws_azure/5007f8c4-afdb-4485-ba19-53	Microsoft.MachineLearningServices	Created
ws_azure/endpoint-score	Machine learning online endpoint	Conflict (Error details)

Errors

Summary

Raw Error

Copy Full Error

TOP ERROR

Resource provider [N/A] isn't registered with Subscription [N/A]. Please see troubleshooting guide, available here: <https://aka.ms/register-resource-provider> (Code: SubscriptionNotRegistered)

WAS THIS HELPFUL?

Explain with Copilot

Troubleshooting Options

Common Deployment Errors

Check Usage and Quotas

New Support Request

Give Feedback

Tell us about your experience with the Errors page

Cloud Deployment Workaround:

Due to limitations in the Azure ML free trial, deployment of the model and generation of the endpoint/API key was not possible. As a workaround:

- The classification model was trained in Jupyter Notebook and saved as a .pkl file
- The saved model was then deployed on Hugging Face Spaces using Gradio, for an interactive web interface for real-time predictions.

Live demo: [Credit Score Classification](#)